

KARTHIK BHARADWAJ

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EDUCATION

Master of Science in Computer Science

Dec. 2027, **GPA: 4.0**

Northeastern University, Boston, MA

Coursework: Algorithms, Database Management Systems, Foundations of AI, Programming Design Paradigms

Bachelor of Engineering in Computer Science and Engineering

2023

R V College of Engineering, Bangalore, India

Coursework: Data Structures & Applications, Operating Systems, Advanced Algorithms, Finite Automata

EXPERIENCE

Digital Associate – Software Engineer

Sept. 2024 – June 2025

Digitide Solutions Ltd, Bangalore, India

- Engineered a modular SaaS platform for AB-InBev serving **1000+** daily users, featuring report generation, an admin control panel, planogram extraction, and bulk update/delete workflows using Angular, Python, Docker, and Azure.
- Developed a chatbot using NLP and Generative AI for warehouse staff to query inventory and planogram data via natural language, reducing manual lookup time by **60%** and eliminating engineering intervention on routine data queries.

Software Engineer – Full Stack

Jan. 2024 – July 2024

zevvo, Bangalore, India

- Engineered core backend systems end-to-end — bookings, billing, subscriptions, and catalog — with MongoDB schema design and JWT auth resistant to XSS/CSRF; built a full admin portal and customer-facing pages, onboarding 50+ active users and reducing auth-related bugs by **40%**
- Automated CI/CD via GitHub Actions deploying Dockerized services to AWS ECS with auto-scaling, boosting deployment frequency by 40%.

Software Development Intern

March 2023 – July 2023

Crestron Electronics, Bangalore, India

- Developed and deployed .NET Core REST APIs with optimized SQL queries, cutting response time by **28%**; built mobile-responsive React components with Google Tag Manager integration to drive behavioral analytics and inform the feature roadmap.

PROJECTS

ParetoFolio – Portfolio Optimization Engine ([Link](#)) | TypeScript, React 19, FastAPI, Python, Pydantic v2

- Built a responsive React 19 frontend with interactive data visualization and a typed FastAPI backend with Pydantic v2 validation, serving a RESTful API for multi-objective portfolio optimization across 75 US equities.
- Implemented NSGA-II entirely from scratch, SBX crossover, polynomial mutation, tournament selection, and crowding distance, producing Pareto-optimal portfolios with an interactive efficient frontier plot and Sharpe-ranked table.

SportsTV Streaming Data Warehouse ([Link](#)) | R, MySQL 8, SQLite, R Markdown, Kimball Star Schema

- Engineered a bulk-batched, idempotent ETL pipeline in R with surrogate-key resolution and partition pruning, achieving 99%+ match rate across 2M+ streaming transactions from heterogeneous sources.
- Modeled a Kimball-style star schema with five conformed dimensions and a RANGE-partitioned fact table on cloud-hosted MySQL 8; generated a fully automated BI report with KPI summaries and device breakdowns driven entirely by live SQL queries.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, .NET core, C#

Backend: Node.js, .NET Core, Express.js, HTTP/REST APIs, Microservices

Frontend: React.js, Next.js, Angular, React Native, HTML/CSS, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, Firebase, MSSQL

Cloud / DevOps: AWS (ECS, S3, Lambda, EC2, ECR), Azure (Container Registry, Blob Storage), GCP, Docker, GitHub Actions, CI/CD

Practices: OOP, System Design (HLD/LLD), Agile/Scrum, Software Development Lifecycle (SDLC), TDD, Code Reviews, Git, API Development, Kubernetes